

Structured, Semi-Structured and Unstructured Data

Aim

To understand and represent the three major types of data — **structured, semi-structured, and unstructured** — using Python with practical examples in DataFrame, JSON, and plain text formats.

Procedure

1. Import the pandas library for handling structured data.
2. Create and display a **structured dataset** using a pandas DataFrame.
3. Define a **semi-structured dataset** using a JSON-like dictionary format and display it.
4. Represent **unstructured data** using plain text to show free-form data without a fixed schema.
5. Compare the three data types to understand their structure and organization.

```
import pandas as pd

structured_data = pd.DataFrame({
    "ID": [1, 2, 3],
    "Name": ["Ismail", "Priya", "Rahul"],
    "Age": [21, 22, 20],
    "Department": ["CSE", "AI", "IT"]
})

print("=== Structured Data Example (DataFrame) ===")
print(structured_data)

semi_structured_data = {
    "students": [
        {"id": 1, "name": "Ismail", "subjects": ["CS", "Math"]},
        {"id": 2, "name": "Priya", "subjects": ["AI", "Data Science"]},
        {"id": 3, "name": "Rahul", "subjects": ["IT", "Networking"]}
    ]
}

print("\n=== Semi-Structured Data Example (JSON) ===")
print(semi_structured_data)

unstructured_data = """
This is an example of unstructured data
"""

print("\n=== Unstructured Data Example (Text) ===")
print(unstructured_data)
```

```
=== Structured Data Example (DataFrame) ===
```

	ID	Name	Age	Department
0	1	Ismail	21	CSE
1	2	Priya	22	AI
2	3	Rahul	20	IT

```
=== Semi-Structured Data Example (JSON) ===
```

```
{'students': [{'id': 1, 'name': 'Ismail', 'subjects': ['CS', 'Math']}, {'id': 2, 'name': 'Priya', 'subjects': ['AI', 'Data Science']}, {'id': 3, 'name': 'Rahul', 'subjects': ['IT', 'Networking']}]}
```

```
=== Unstructured Data Example (Text) ===
```

```
This is an example of unstructured data
```

Result

Hence, a Python program to demonstrate **structured, semi-structured, and unstructured data** has been successfully implemented, clearly distinguishing between tabular, hierarchical, and free-text data representations.